SMART INDIA HACKATHON 2026

TITLE PAGE

- **Problem Statement ID –** 26079
- **Problem Statement Title-** AI-Based Forecast Bust Detection for Medium-Range Weather Forecasts
- **Theme-** Smart Automation
- **PS Category- Software**
- **Team ID- 196700**
- **Team Name -** Team apex001

AI-BASED FORECAST BUST DETECTION

SMART INDIA HACKATHON 2026

CORE SOLUTION: AI-BASED FORECAST BUST DETECTION

1. 

Data Collection: Collect observed and medium-range forecast weather data.

2. 

Data Processing: Identify and clean data, noise, and abnormal values.

3. 

AI Forecast Comparison: AI compares Weather Forecast vs Actual/Observed Weather.

4. 

Forecast Error Detection: Calculate forecast error in temperature, rainfall, wind, pressure, etc.

5. 

Forecast Bust Detection: AI detects large and unusual errors to classify as Normal / High Error / Forecast Bust.

6. 

Confidence & Risk Score: Generate Confidence Score and Risk Level for each forecast.

7. 

Early Warning & Dashboard: Show alerts and visualizations of detected forecast busts on a dashboard.

CORE SYSTEM WORKFLOW

INNOVATION & UNIQUENESS



AI-Based Early Bust Detection



Forecast vs Actual Analysis



Confidence & Risk Scoring



Real-Time Alerts & Visualization



Decision Support for Agriculture, Disaster Management & Other Sectors

INNOVATION & UNIQUENESS



Direct Connection
Farmer ↔ Buyer



Market Intelligence
AI + Real Data



Verified Buyers
Trusted & Safe



Transparent Information
Fair Prices for All

TECHNICAL FLOW

SMART INDIA HACKATHON 2026

1 Weather Data Sources

- Satellite Data
- Radar Data
- Weather Stations
- Weather APIs (e.g., OpenWeather, IMD, etc.)

2 Data Collection

Historical Weather + Medium-Range Forecast Data

- Past observations
- Forecast data
- Multiple time ranges

3 Data Preprocessing & Quality Check

- Missing value handling
- Noise removal
- Data normalization
- Outlier detection

4 Feature Extraction

Temperature
Rainfall
Wind Speed
Pressure
Humidity
etc.

Convert raw data into useful features for AI model.

5 AI/ML Model

Forecast vs Actual Weather Comparison

- Machine Learning
- Deep Learning
- Time Series Analysis

6 Error & Anomaly Detection

- Calculate forecast error
- Detect unusual patterns
- Identify anomaly

7 Forecast Bust Detection

Normal (Within range)

High Error (Warning)

Forecast Bust (Severe Error)

8 Confidence Score Generation

78%

- Model confidence
- Error level
- Risk probability

9 Alert & Visualization Dashboard

- Real-time alerts
- Maps & charts
- Forecast vs Actual view
- Bust status

10 Early Warning & Decision Support

- Government agencies
- Disaster management
- Agriculture
- Other sectors

Input

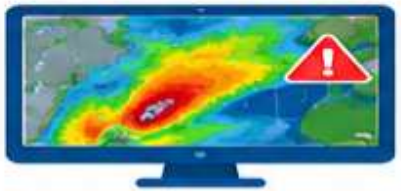
Forecast + Historical / Observed Weather Data

- Weather parameters
- Medium-range forecast
- Past observations

Output

Forecast Bust Detection

- ✓ Normal / High Error / Forecast Bust
- ✓ Confidence Level
- ✓ Early Warning Alerts



Better Forecasts → Less Risk → Safer Future

FEASIBILITY & VIABILITY

SMART INDIA HACKATHON 2026

FEASIBILITY

Technical Feasibility:
- Weather data + AI/ML models se forecast errors detect karna

Data Availability:
- Historical weather data aur actual observations ka use

Operational Feasibility:
- Forecast ko automatically evaluate, weather aur compare, Bust Alert generate karna

VIABILITY

Economic Viability:
- Existing weather datasets aur cloud infrastructure se development cost manageable

Scalability:
- System ko different regions aur weather conditions mein expand karna

Real-World Impact:
- Agriculture, disaster management, aviation aur weather-sensitive sectors ko better decision support

SYSTEM CHALLENGES & MITIGATIONS

1. **Model Uncertainty**
   Mitigation → Ensemble modeling & confidence scoring
2. **Compute Constraints**
   Mitigation → Cloud auto-scaling & optimized pipelines
3. **Alert Precision**
   Mitigation → Threshold tuning & validation metrics
4. **Data Volume**
   Mitigation → Efficient DB & data pruning strategies
5. **Interpretability**
   Mitigation → Explainable AI (SHAP/LIME) integration
6. **Latency**
   Mitigation → Real-time streaming & edge processing

IMPLEMENTATION PLAN

1. **Pilot Phase** (Select Regions)
   • Selected districts for initial testing & model validation
2. **Expansion** (Regional Scale)
   • Integration across multiple states & regional data inputs
3. **Full Deployment** (Pan-India)
   • Nationwide rollout with monitoring dashboard & feedback loop

IMPACT AND BENEFITS

SMART INDIA HACKATHON 2026

Impacts

Forecast Reliability Improve
Weather forecast ki reliability aur accuracy badhegi.

Early Detection of Errors
AI forecast mein major errors ko pehle detect karega.

Better Disaster Preparedness
Flood, heavy rainfall, heatwave jaise events ke liye preparation better hogi.

Improved Decision Making
Government aur weather agencies ko timely decisions lene mein help milegi.

Reduced Weather Risk
Wrong forecast ke wajah se hone wale risks ko kam kiya ja sakega.

Benefits for Sectors

Agriculture	Farmers ko farming activities plan karne mein help milegi.
Disaster Management	Emergency teams ko timely alerts aur better planning milegi.
Transport & Aviation	Weather-related disruptions aur risks ko reduce karne mein help.
Energy Sector	Solar, wind aur electricity demand planning better hogi.
Economic Savings	Weather-related financial losses ko reduce karne mein help.
Public Safety	Log severe weather conditions ke liye pehle se prepared rahenge.

RESEARCH AND REFERENCES

Data & Institutional Sources

ECMWF – Medium-Range Weather Forecasts
Up to 15-day weather forecasts and ensemble prediction.
ECMWF Medium-Range Forecasts

ECMWF – Forecast Uncertainty & Ensemble Prediction
Forecast uncertainty and ensemble-based risk estimation.
ECMWF Forecast Uncertainty

WMO – Artificial Intelligence in Weather and Climate
AI applications in weather forecasting and early-warning systems.
WMO Artificial Intelligence

ECMWF – Artificial Intelligence Forecasting System (AIFS)
ECMWF's operational AI-based weather forecasting system.
ECMWF AIFS

GraphCast – Google DeepMind
AI-based global medium-range weather forecasting for up to 10 days.
GraphCast Research – Google DeepMind

Pangu-Weather – AI Weather Forecasting
AI-based global weather forecasting research.
Pangu-Weather – Nature

Research Areas Drawn Upon

Artificial Intelligence & Machine Learning
Weather Forecasting & Numerical Weather Prediction (NWP)
Ensemble Forecasting & Uncertainty Quantification
Time-Series Analysis
Forecast Error & Anomaly Detection
Deep Learning for Weather Prediction
Historical Weather & Climate Data Analysis
Predictive Analytics & Risk Scoring
AI-Based Decision Support Systems
Early Warning & Disaster Risk Reduction